PROD

CSRSS.EXE OUTSIDE SYSTEM32

Detect masqueraded Windows Client/Server Runtime execution from non-system paths, observed in the NIK C2 implant.

CLASSIFICATION

Production candidate - Untested implementation sketch. Confidence Medium. Behavioral masquerading signal.

PRIMARY PLATFORM

Microsoft Defender XDR. Required telemetry: DeviceProcessEvents.

OPERATIONAL VALUE 9/10

A process named csrss.exe outside the legitimate System32 path is highly specific, durable beyond NIK C2, and immediately investigable.

Executive Assessment

Facts, interpretation and explicit boundaries

SOURCE FACT

Cisco Talos observed NIK C2 samples in the wild and assessed with moderate confidence that the framework is functional, not merely a proof of concept.

SOURCE FACT

At startup the implant renames itself to csrss.exe inside the Windows temporary directory and applies a critical-process flag.

ANALYST INFERENCE

The executable-name and path mismatch is a durable process-masquerading signal. It detects NIK C2 and unrelated abuse using the same protected system-process name.

NOT PROVEN BY A MATCH

NIK C2 identity, AI API use, credential theft, C2 success, ransomware deployment, privilege escalation or actor attribution.

Observed NIK C2 Flow

Only stages directly supported by Talos

1

Execution mode

Arguments select primary agent, watchdog or copy-watchdog behavior.

2

Masquerade

The implant uses csrss.exe or simvol.exe and places a csrss.exe copy in Windows Temp.

3

Persistence

The implant copies into AppData Roaming, configures Run-key autostart and starts a watchdog.

4

C2

TLS WebSocket on 443 operates alongside 250 ms HTTP fallback polling.

5

Post-compromise

Commands support credential theft, surveillance, proxying, evasion and screen locking.

DETECTION BOUNDARY

The production candidate implements only csrss.exe process execution outside the canonical System32 location.

Telemetry & Fields

Documented Defender XDR process-event schema

DeviceProcessEvents	**Required**	Process creation telemetry
FileName	**Required**	Case-insensitive csrss.exe identity
FolderPath	**Required**	Full executable path
DeviceId	**Required**	Endpoint correlation key
ProcessUniqueId	**Enrichment**	Strong process pivot
SHA1	**Enrichment**	File provenance pivot
InitiatingProcess*	**Enrichment**	Parent and launch context

PATH ASSUMPTION

Legitimate Windows csrss.exe is expected under the Windows System32 directory. Validate Windows-directory variants and legacy systems before alerting.

Primary Detection Candidate

Microsoft Defender XDR - system-process path mismatch

```
// Production candidate - Untested implementation sketch

DeviceProcessEvents

| where FileName =~ "csrss.exe"

| where not(FolderPath startswith @"C:\Windows\System32\")

| project Timestamp, DeviceId, DeviceName, AccountName,

          FileName, FolderPath, SHA1, ProcessCommandLine,

          ProcessIntegrityLevel, ProcessTokenElevation,

          InitiatingProcessFileName, InitiatingProcessFolderPath,

          InitiatingProcessCommandLine, ProcessId,

          ProcessUniqueId, ReportId

| order by Timestamp desc
```

CONDITION EVIDENCE

Talos observed csrss.exe in Windows Temp. FileName and FolderPath are documented fields. The exclusion encodes the legitimate canonical path; no IOC, threshold, join or rarity is used.

STATUS

Production candidate - Untested implementation sketch. Validate path normalization and benign exceptions before enabling an alert.

Cross-SIEM Assessment

No mechanical translations

SPLUNK

No reliable implementation can currently be written. Required: endpoint process telemetry, index, sourcetype, parser and verified executable-path field extraction.

CHRONICLE YARA-L 2.0

No reliable implementation can currently be written. Required: parser name and verified UDM mapping for target process file name and full path.

MICROSOFT SENTINEL

The primary query targets Defender XDR Advanced Hunting. If MDE streaming is ingested into Log Analytics, verify the connector and table semantics rather than assuming parity.

EVIDENCE TO COLLECT

Any non-System32 csrss.exe execution, file signer and hash, parent process, file origin, related Run-key activity, network connections and host containment state.

Validation & Tuning Plan

Promotion requires environmental path verification

01 Run across available retention and inventory every distinct FolderPath for csrss.exe.

02 Confirm expected Windows path representation across supported OS versions, language packs and redirected Windows directories.

03 For each result, validate SHA1, certificate trust, file origin, parent process and ProcessUniqueId pivots.

04 Review related registry, file and network events; NIK C2 attribution requires corroborating evidence.

05 Reject or redesign if legitimate software commonly executes a private csrss.exe copy outside System32.

SUCCESS CRITERION

All legitimate csrss.exe executions resolve to canonical system paths, while simulated or recovered masqueraded copies produce complete, investigable events.

ATT&CK, Conclusion & References

Evidence-backed scope and publication record

MITRE ATT&CK T1036.005

The directly supported behavior is masquerading with a legitimate Windows process name from a non-canonical location: Match Legitimate Resource Name or Location.

CONCLUSION

Non-System32 csrss.exe execution is a compact, high-signal production candidate. A match warrants immediate validation but does not alone establish NIK C2.

PRIMARY RESEARCH

Cisco Talos. NIK C2: The new AI-enabled command-and-control framework. 29 October 2026.

hxxps://blog[.]talosintelligence[.]com/nik-c2-new-ai-enabled-framework/

OFFICIAL SCHEMA

Microsoft Learn. DeviceProcessEvents advanced hunting table.

hxxps://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table

QUALITY CONTROLS

Eight A4 pages; no arbitrary threshold; no customer data; links defanged; no invented cross-SIEM mappings.